

PART A — QUESTIONS

AF13-001

Question:

Solve: $15x - 12 = 78$

- A) 5
- B) 6
- C) 7
- D) 8

AF13-002

Question:

If $13n + 9 = 100$, what is n ?

- A) 6
- B) 7
- C) 8
- D) 9

AF13-003

Question:

Simplify: $6(3x - 2) - 8x$

- A) $10x - 12$
- B) $18x - 12$
- C) $10x - 2$
- D) $26x - 12$

AF13-004

Question:

A rental company charges \$24 per hour plus a \$36 service fee. Which expression represents the total cost for h hours?

- A) $24h + 36$
- B) $36h + 24$
- C) $60h$
- D) $36 - 24h$

AF13-005

Question:

If $f(x) = 10x - 17$, what is $f(7)$?

- A) 43
- B) 53
- C) 70
- D) 87

AF13-006

Question:

Which ordered pair satisfies the equation $y = 8x + 1$?

- A) (1, 8)
- B) (2, 16)
- C) (3, 25)
- D) (4, 34)

AF13-007

Question:

Solve: $14(x - 5) = 84$

- A) 9
- B) 10
- C) 11
- D) 12

AF13-008

Question:

A pattern begins: 4, 16, 28, 40, ... What is the next number?

- A) 48
- B) 50
- C) 52
- D) 54

AF13-009

Question:

Solve: $x/22 = 4$

- A) 26
- B) 66
- C) 88
- D) 96

AF13-010

Question:

A line has a slope of -10 and crosses the y-axis at 7. Which equation represents the line?

- A) $y = 7x - 10$
- B) $y = -10x + 7$
- C) $y = 10x + 7$
- D) $y = -7x + 10$

AF13-011

Question:

If 8 cases contain 184 parts, how many parts are in 5 cases at the same rate?

- A) 105
- B) 110
- C) 115
- D) 120

AF13-012

Question:

Solve: $90 - 14x = 34$

- A) 3
- B) 4
- C) 5
- D) 6

AF13-013

Question:

Which expression is equivalent to $45x - 18x + 10$?

- A) $27x + 10$
- B) $63x + 10$
- C) $27x - 10$
- D) $45x - 8$

AF13-014

Question:

A machine starts with 240 completed parts and produces 16 more parts each hour. Which equation gives the total number of parts P after h hours?

- A) $P = 240h + 16$
- B) $P = 16h + 240$
- C) $P = 256h$
- D) $P = 240 - 16h$

AF13-015

Question:

If $y = -12x + 110$, what is y when $x = 8$?

- A) 14
- B) 22
- C) 96
- D) 206

AF13-016

Question:

Solve: $20x - 18 = 12x + 54$

- A) 7
- B) 8
- C) 9
- D) 10

AF13-017

Question:

Which value of x makes $11x - 4$ less than 84?

- A) 7
- B) 8
- C) 9
- D) 10

AF13-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : -6, 7, 20, 33

What is the rule?

- A) $y = 13x - 6$
- B) $y = 6x + 13$
- C) $y = -6x + 13$
- D) $y = 13x + 6$

AF13-019

Question:

Solve: $10(x + 3) = 7x + 57$

- A) 7
- B) 8
- C) 9
- D) 10

AF13-020

Question:

If $a = -8$ and $b = 6$, what is $4a + 5b$?

- A) -62
- B) -2
- C) 2
- D) 62

AF13-021

Question:

Which point is located in Quadrant III?

- A) (9, -7)
- B) (-9, 7)
- C) (-9, -7)
- D) (9, 7)

AF13-022

Question:

A worker packages 132 items in 11 minutes. At the same rate, how many items can the worker package in 8 minutes?

- A) 84
- B) 90
- C) 96
- D) 108

AF13-023

Question:

Solve for w : $Q = 16w - 32$

- A) $w = Q + 32$
- B) $w = (Q + 32)/16$
- C) $w = 16(Q + 32)$
- D) $w = Q/16 - 32$

AF13-024

Question:

What is the slope of a line that goes through (5, 12) and (11, 54)?

- A) 6
- B) 7
- C) 8
- D) 42

AF13-025

Question:

Simplify: $31x - 14 - 22x + 25$

- A) $9x + 11$
- B) $53x + 11$
- C) $9x - 39$
- D) $53x - 39$

AF13-026

Question:

If $g(x) = x^2 + 4x$, what is $g(9)$?

- A) 81
- B) 99
- C) 117
- D) 121

AF13-027

Question:

A sequence follows the rule “multiply by 2, then subtract 9.” If the first term is 15, what is the second term?

- A) 21
- B) 24
- C) 30
- D) 39

AF13-028

Question:

Solve: $13(x - 4) + 11 = 76$

- A) 7
- B) 8
- C) 9
- D) 10

AF13-029

Question:

A line crosses the y-axis at 14 and falls 7 units for every 1 unit to the right. Which equation represents the line?

- A) $y = 7x + 14$
- B) $y = -7x + 14$
- C) $y = 14x - 7$
- D) $y = -14x + 7$

AF13-030

Question:

If $15/20 = x/60$, what is x ?

- A) 35
- B) 40
- C) 45
- D) 50

AF13-031

Question:

Solve: $-15x + 45 = -90$

- A) -9
- B) -6
- C) 6
- D) 9

AF13-032

Question:

A supply shelf starts with 1,500 washers. Each kit uses 75 washers. Which expression gives the number of washers left after k kits?

- A) $1,500 + 75k$
- B) $1,500 - 75k$
- C) $75k - 1,500$
- D) $1,500 \div 75k$

AF13-033

Question:

Which equation has $x = 18$ as its solution?

- A) $2x + 12 = 50$
- B) $5x - 14 = 76$
- C) $x/6 + 7 = 11$
- D) $4x - 9 = 60$

PART B — ANSWER KEY + EXPLANATIONS

AF13-001 — Correct: B — Explanation:

Solve $15x - 12 = 78$. Add 12 to both sides: $15x = 90$. Divide by 15: $x = 6$. The most common mistake is subtracting 12 instead of adding 12.

AF13-002 — Correct: B — Explanation:

Start with $13n + 9 = 100$. Subtract 9 from both sides: $13n = 91$. Divide by 13: $n = 7$. A common mistake is stopping at 91 and forgetting to divide by 13.

AF13-003 — Correct: A — Explanation:

Distribute first: $6(3x - 2) = 18x - 12$. Then subtract $8x$: $18x - 8x - 12 = 10x - 12$. A common mistake is forgetting to multiply -2 by 6.

AF13-004 — Correct: A — Explanation:

The rental company charges \$24 for each hour, so that part is $24h$. The \$36 service fee is added one time. The total cost is $24h + 36$. A common mistake is treating the service fee as the hourly rate.

AF13-005 — Correct: B — Explanation:

Substitute 7 for x : $f(7) = 10(7) - 17 = 70 - 17 = 53$. A common mistake is adding 17 instead of subtracting it.

AF13-006 — Correct: C — Explanation:

Test the choices in $y = 8x + 1$. For $(3, 25)$, $8(3) + 1 = 24 + 1 = 25$, which matches the y -value. A common mistake is choosing a point that is close but does not make the equation true.

AF13-007 — Correct: C — Explanation:

Solve $14(x - 5) = 84$. Divide both sides by 14: $x - 5 = 6$. Add 5: $x = 11$. A common mistake is adding 5 before undoing the multiplication by 14.

AF13-008 — Correct: C — Explanation:

The pattern increases by 12 each time: 4, 16, 28, 40. Add 12 to 40: 52. A common mistake is assuming the increase is 10 because the numbers are large.

AF13-009 — Correct: C — Explanation:

$x/22 = 4$ means x divided by 22 equals 4. Multiply both sides by 22: $x = 88$. A common mistake is adding 22 and 4 instead of multiplying.

AF13-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -10 and the y-intercept is 7, so the equation is $y = -10x + 7$. A common mistake is switching the slope and y-intercept.

AF13-011 — Correct: C — Explanation:

Find the unit rate: $184 \text{ parts} \div 8 \text{ cases} = 23 \text{ parts per case}$. For 5 cases: $23 \times 5 = 115$. A common mistake is multiplying 184 by 5 without finding the rate first.

AF13-012 — Correct: B — Explanation:

Solve $90 - 14x = 34$. Subtract 90 from both sides: $-14x = -56$. Divide by -14: $x = 4$. A common mistake is losing the negative sign after subtracting 90.

AF13-013 — Correct: A — Explanation:

Combine like terms: $45x - 18x = 27x$. The constant 10 stays the same. The expression becomes $27x + 10$. A common mistake is combining the constant with the variable terms.

AF13-014 — Correct: B — Explanation:

The machine starts with 240 completed parts. It produces 16 more parts each hour. After h hours, the total is $P = 16h + 240$. A common mistake is making 240 the hourly rate.

AF13-015 — Correct: A — Explanation:

Substitute $x = 8$ into $y = -12x + 110$: $y = -12(8) + 110 = -96 + 110 = 14$. A common mistake is treating -12(8) as positive 96.

AF13-016 — Correct: C — Explanation:

Solve $20x - 18 = 12x + 54$. Subtract $12x$: $8x - 18 = 54$. Add 18: $8x = 72$. Divide by 8: $x = 9$. A common mistake is adding 18 to only one side.

AF13-017 — Correct: A — Explanation:

Solve $11x - 4 < 84$. Add 4: $11x < 88$. Divide by 11: $x < 8$. The only listed value less than 8 is 7. A common mistake is choosing 8, but $11(8) - 4 = 84$, not less than 84.

AF13-018 — Correct: A — Explanation:

The y-values increase by 13 each time x increases by 1, so the slope is 13. When $x = 0$, $y = -6$, so the y-intercept is -6. The rule is $y = 13x - 6$. A common mistake is using +6 instead of -6.

AF13-019 — Correct: C — Explanation:

Distribute: $10x + 30 = 7x + 57$. Subtract $7x$: $3x + 30 = 57$. Subtract 30: $3x = 27$. Divide by 3: $x = 9$. A common mistake is forgetting to multiply 3 by 10.

AF13-020 — Correct: B — Explanation:

Substitute $a = -8$ and $b = 6$: $4a + 5b = 4(-8) + 5(6) = -32 + 30 = -2$. A common mistake is treating $4(-8)$ as positive 32.

AF13-021 — Correct: C — Explanation:

Quadrant III has a negative x-coordinate and a negative y-coordinate. The point $(-9, -7)$ fits this description. A common mistake is choosing $(-9, 7)$, which is Quadrant II.

AF13-022 — Correct: C — Explanation:

132 items in 11 minutes means $132 \div 11 = 12$ items per minute. In 8 minutes: $12 \times 8 = 96$. A common mistake is multiplying 132 by 8 without finding the rate first.

AF13-023 — Correct: B — Explanation:

Start with $Q = 16w - 32$. Add 32 to both sides: $Q + 32 = 16w$. Divide by 16: $w = (Q + 32)/16$. A common mistake is dividing Q by 16 first and then adding 32.

AF13-024 — Correct: B — Explanation:

Slope equals change in y divided by change in x. From $(5, 12)$ to $(11, 54)$, y increases by 42 and x increases by 6. The slope is $42 \div 6 = 7$. A common mistake is using 42 as the slope without dividing by the change in x.

AF13-025 — Correct: A — Explanation:

Combine like terms: $31x - 22x = 9x$, and $-14 + 25 = 11$. The simplified expression is $9x + 11$. A common mistake is writing $9x - 11$ because the positive 25 is missed.

AF13-026 — Correct: C — Explanation:

Substitute 9 for x: $g(9) = 9^2 + 4(9) = 81 + 36 = 117$. A common mistake is calculating 9^2 as 18 instead of 81.

AF13-027 — Correct: A — Explanation:

Start with 15. Multiply by 2: 30. Then subtract 9: 21. A common mistake is subtracting 9 first and then multiplying.

AF13-028 — Correct: C — Explanation:

Solve $13(x - 4) + 11 = 76$. Distribute: $13x - 52 + 11 = 76$. Combine: $13x - 41 = 76$. Add 41: $13x = 117$. Divide by 13: $x = 9$. A common mistake is forgetting to combine -52 and $+11$.

AF13-029 — Correct: B — Explanation:

“Falls 7 units for every 1 unit to the right” means the slope is -7 . The line crosses the y-axis at 14, so the equation is $y = -7x + 14$. A common mistake is using a positive slope because the number 7 is given.

AF13-030 — Correct: C — Explanation:

Use the proportion $15/20 = x/60$. Since $20 \times 3 = 60$, multiply 15 by 3: $x = 45$. A common mistake is adding 40 to the numerator because 20 becomes 60.

AF13-031 — Correct: D — Explanation:

Solve $-15x + 45 = -90$. Subtract 45 from both sides: $-15x = -135$. Divide by -15 : $x = 9$. A common mistake is giving -9 because both sides contain negative numbers.

AF13-032 — Correct: B — Explanation:

The shelf starts with 1,500 washers. Each kit uses 75 washers, so k kits use 75k washers. The number left is $1,500 - 75k$. A common mistake is adding 75k even though washers are being used.

AF13-033 — Correct: B — Explanation:

Test $x = 18$. In choice B, $5(18) - 14 = 90 - 14 = 76$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.